

JPE Replication Report

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1 Paper Identification

ID	JPE-Richert-20240555
Author	Richert, Eric
Title	Quantity Commitments in Multiunit Auctions: Evidence from Credit Event Auctions
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

- We found quite a few bugs in your code. Please check, and re-run on a different computer before resubmitting.
- Please address all items marked as **{REQUIRED}**.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)

- ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
 - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

- ☐ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
immspreads_clean.csv	no	yes	yes	yes	yes	yes
openinterest_clean.csv	no	yes	yes	yes	yes	yes
limitorders_clean.csv	no	yes	yes	yes	yes	yes
auctionlistid.csv	no	yes	yes	yes	yes	yes
bondtypes.csv	no	yes	yes	yes	yes	yes
gosyop23q6grk5y1.csv	no	yes	yes	yes	yes	yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - MATLAB R2023a
 - Toolboxes: Statistics and Machine Learning Toolbox, Optimization Toolbox, Parallel Computing Toolbox
- ☒ Computational Requirements specified as follows:
 - Hardware: 20-core cluster node or a standard multi-core desktop
 - Storage: Sufficient space for large intermediate .mat files (up to ~700 MB for smc_cfs_np.mat)
- ☒ Time Requirements specified as follows:

- approximately 20 hours on a 20-core cluster node, or 3–4 days on a desktop

✓ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this – please check!

Summary: - Data files with PII indicators: 5 - Variables flagged in data: 6 - Code files with PII references: 54 - PII references in code: 607

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	<code>auctionlistid.csv</code>	1	name
Data	<code>bondtypes.csv</code>	2	name, loc
Data	<code>immspreads_clean.csv</code>	1	name
Data	<code>limitorders_clean.csv</code>	1	name
Data	<code>openinterest_clean.csv</code>	1	name
Code	<code>BsplineBasis3.m</code>	24	city, second, son, degree
Code	<code>BsplineEval3.m</code>	1	second
Code	<code>acrossrounds.m</code>	1	second
Code	<code>basic_estimator.m</code>	56	lon, lat, city
Code	<code>basic_estimator_copypdrop.m</code>	56	lon, lat, city
Code	<code>bondpriceimport.m</code>	14	name, loc
Code	<code>bondprices.m</code>	1	lat

File Type	File	Variables/References	PII Categories
Code	<code>build_bondforestimation.m</code>	2	name
Code	<code>calculate_surplusraw.m</code>	5	lon, son
Code	<code>cdf_estimator.m</code>	1	lat
Code	<code>cdf_estimator_imm.m</code>	2	lat
Code	<code>complex_bootstrap.m</code>	6	second, lon, lat
Code	<code>complex_estimator_step1.m</code>	71	lon, lat, city
Code	<code>complex_estimator_step2.m</code>	12	lon, lat
Code	<code>complex_estimator_step2b.m</code>	11	lon, lat
Code	<code>data_summary.m</code>	18	name, block, loc, lat, second
Code	<code>debug_bondforestimation.m</code>	3	name
Code	<code>doubleauction0uter_1stepspecial_yin.m</code>	3	lat
Code	<code>dscatter.m</code>	11	loc, location, lon, name, second
Code	<code>emcdf.m</code>	1	lat
Code	<code>firststagebidding.m</code>	10	loc, lat
Code	<code>heatscatter.m</code>	9	name, lat
Code	<code>jacobianest.m</code>	13	loc, location, lat, second
Code	<code>job0_validate.m</code>	5	name, loc
Code	<code>job0_validate.sh</code>	1	name
Code	<code>job1_dataprep.m</code>	1	name
Code	<code>job1_dataprep.sh</code>	1	name
Code	<code>job2a_estimation.m</code>	3	loc, location, name
Code	<code>job2a_estimation.sh</code>	1	name

File Type	File	Variables/References	PII Categories
Code	job2b_estimation.m	3	loc, location, name
Code	job2b_estimation.sh	1	name
Code	job3a_postcfs.m	4	loc, location, lat, name
Code	job3a_postcfs.sh	1	name
Code	job3b_postcfs.m	3	loc, location, name
Code	job3b_postcfs.sh	1	name
Code	job3c_postcfs.m	3	loc, location, name
Code	job3c_postcfs.sh	1	name
Code	main_cds.m	3	lat, loc, location, name
Code	normalize_prices.m	28	lon, lat, second, name
Code	npestimator.m	20	lon
Code	permn.m	4	son, name, lat, second
Code	pmcalculation.m	1	lat
Code	postestimation_clean.m	14	lat, lon, son, url
Code	postmain_cfs.m	1	lat
Code	pricechangebs.m	1	loc, location
Code	robustnesschecks.m	6	lat, lon
Code	round1_quotescalibration.m	8	lat, lon
Code	smc_cfs_yin.m	30	lat, block, loc
Code	table2latex.m	20	lat, name, son
Code	truthfulpimmcheck.m	3	lat
Code	v_correction.m	3	lon

File Type	File	Variables/References	PII Categories
Code	weightedMedian.m	4	lat
Code	weightedcorrs.m	100	lat, son, second, loc, location
Code	weightsolnpricespartialgridINTs1_yin.m	1	lon

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-03-23T11:46:15.483

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for all STATA applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
80	9	29	15	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/CDS/confidential-data-not-for-publication/gosyop23q6grk5y1.csv	129.31

7.2 Code description

There are provided 37 MATLAB, 1 of which is the master script (`code/main_cds.m`).

cloc		github.com/AIDanial/cloc v 2.06 T=0.34 s (250.1 files/s, 75823.1 lines/s)		
Language	files	blank	comment	code
CSV	5	0	0	14416
MATLAB	73	1153	2109	7732
Markdown	1	153	0	320
Bourne Shell	7	22	98	69
---	---	---	---	---
SUM:	86	1328	2207	22537

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

Hardware:

- Nuvolos: Ubuntu 24.04.1 LTS, 4 vCPU + 16 GB RAM

Software:

- Matlab R2025b

8 Replication

- Downloaded code and data from dropbox.
- Copied git repo and unzipped the replication package into this repo.
- Set up the computing environment on Nuvolos and scaled the hardware to 16 vCPU and 64 GB RAM.
- Set working directory and ran the code as per README.
- Modified `code/main_cds.m` to set `ncores = 16` to match the available cluster hardware as instructed in the `README.md`
- Ran the code as per the README, but code broke executing the Step 1 caused by line 68 of `bondpriceimport.m`

Error using :
Colon operands must be real scalars.


```
Error in bondpriceimport (line 68)
    for jk=1:size(bonddlocs)
        ^
```

```
Error in main_cds (line 58)
bondpriceimport
^^^^^^^^^^^^^^
```

- Committed the repo at this exact broken state to preserve the generated error log file
- Substituted this bit of code on line 68 of `bondpriceimport.m`:

```
for jk=1:size(bonddlocs)
```

- With this:

```
for jk=1:length(bonddlocs)
```

- Ran code and got a new error:

```
Error using :
Colon operands must be real scalars.
```

```
Error in acrossrounds (line 43)
for aa=1:size(aucidfslist)
    ^
```

```
Error in firststagebidding (line 89)
acrossrounds
^^^^^^^^^^^^
```

```
Error in data_summary (line 97)
firststagebidding
^^^^^^^^^^^^^^^^
```

```
Error in main_cds (line 65)
    data_summary
    ^^^^^^^^^^^^^
```

- Committed changes with the new log output. Then, proceeded with substituting the bit of code on line 43 of `acrossrounds.m`:

```
for aa=1:size(aucidfslist)
```

- With this:

```
for aa=1:length(aucidfslist)
```

- Hit another error:

```
Error using :
Colon operands must be real scalars.
```

```
Error in data_summary (line 240)
    for ia=1:size(tempBlist)
```

```

      ^
Error in main_cds (line 65)
  data_summary
  ~~~~~

```

- Committed changes with the new log output. Then, proceeded with substituting the bit of code on line 240 of `data_summary.m`:

```
for ia=1:size(tempBlist)
```

- With this:

```
for ia=1:length(tempBlist)
```

- Error seems to be systematic in all scripts. Aborted correction and started evaluating output obtained so far after this error message:

```

Error using :
Colon operands must be real scalars.

```

```

Error in data_summary (line 256)
for ia=1:size(tempBlist)
^

```

```

Error in main_cds (line 65)
  data_summary
  ~~~~~

```

- Committed changes

9 Findings

Error message caused by line 68 of `bondpriceimport.m`

After fix, error message cause by line 43 of `acrossrounds.m`.

After fix, error message cause by line 240 and then 256 of `data_summary.m`.

Probably, other scripts present the same issue. See committed log for more details.

9.1 Data Preparation Code

- Program `main_cds.m` stopped after encountering error. Output not as expected.

9.2 Tables and Figures

Figure/Table #	Program	Line Number	Reproduced?
Figure 1A	normalize_prices.m	458	No
Figure 1B	normalize_prices.m	469	No
Figure 3	/		No
Figure 4 (left)	postestimation_clean.m	395	No
Figure 4 (middle)	postestimation_clean.m	408	No
Figure 4 (right)	postestimation_clean.m	416	No
Table 1	data_summary.m	130	Yes
Table 2	data_summary.m	158	Yes
Table 3	normalize_prices.m	263	No
Table 4	postestimation_clean.m	219	No
Table 5	postmain_cfs.m	76	No
Table A.1	normalize_prices.m	437	No
Table OS.1	data_summary.m	77	Yes
Table OS.2	data_summary.m	305	No
Table OS.3	data_summary.m	200	Yes
Table OS.4	normalize_prices.m	326	No
Table OS.5	normalize_prices.m	379	No
Table OS.6	main_cds.m	153	No
Figure OS.3	bondprices.m	69	Yes
Figure OS.4	data_summary.m	356	No
Figure OS.5	data_summary.m	334	No
Figure OS.6	robustnesschecks.m	25, 31	No
Figure OS.7	firststagebidding.m	97	Yes
Figure OS.8	postestimation_clean.m	330	No

- Figure OS3: Looks the same but the scaling in the reproduced figure is not 1:1 for both axes.

- Figure OS7: Looks the same but the scaling in the reproduced figure is not 1:1 for both axes as in the original.
- Table 1: output looks different.

ID	Dealer	Bid/offer	Size (\$M)
1	Barclays Bank PLC	Offer	10
2	BNP Paribas SA	-	0
3	Credit Suisse	Offer	7.953
4	Deutsche Bank	-	0
5	Goldman Sachs International	Bid	6.53
6	J.P. Morgan Securities LLC.	Offer	26.974
7	Merrill Lynch, Pierce, Fenner & Smith Inc.	-	0
8	Morgan Stanley & Co. LLC	Offer	9.0
9	Societe Generale	-	0
Total for Auction (Net Open Interest)		Offer	47.397

Paper version

Table 1: Initial Stage Quantities (Parker Drilling)

Dealer	Bid/Offer	Size (\$M)
bnpparibas	Offer	9.000
barclays	Offer	26.974
societegenerale	—	0.000
merrillynch	—	0.000
goldmansachs	—	0.000
morganstanley	Offer	7.953
deutschebank	—	0.000
jpmorgan	Offer	10.000
suisse	Bid	6.530
Net Open Interest	Offer	47.397

Replicated version

Figure 1: Table 1

- Table 2: output looks different and some results are missing.

Name	Submissions			Sorted Bids		Sorted Offers	
	ID	Bid	Offer	ID	Bid	ID	Offer
Barclays Bank PLC	1	53	58	5	56	8	55
BNP Paribas SA	2	54	59	2	54	1	58
Credit Suisse	3	53	58	4	53.5	3	58
Deutsche Bank	4	53.5	58.5	9	53.5	6	58
Goldman Sachs International	5	56	61	3	53	7	58
JP Morgan Securities LLC	6	53	58	6	53	4	58.5
Merrill Lynch, Pierce, Fenner & Smith Inc	7	53	58	7	53	9	58.5
Morgan Stanley & Co., LLC	8	50	55	1	53	2	59
Societe Generale	3	53.5	58.5	8	50	5	61

Paper version

Table 2: Initial Stage Price Quotes (Parker Drilling)

Dealer	Bid	Offer
bnpparibas	54.00	59.00
barclays	53.00	58.00
societegenerale	53.50	58.50
merrilllynch	53.00	58.00
goldmansachs	55.00	60.00
morganstanley	51.00	56.00
deutschebank	53.50	58.50
jpmorgan	53.00	58.00
suisse	53.00	58.00
IMM = 55.75		

Replicated version

Figure 2: Table 2

- Table OS.1: output looks different and some results are missing.

	Mean	Sd	[P10,P90]
N Bonds	10.77	27.70	[1,16.90]
Max maturity (years)	10.42	11.93	[0,26.9]
Min maturity (years)	2.97	4.92	[0.6,90]
Max coupon %	5.5	4.8	[0,10.75]
Min coupon %	2.7	3.3	[0,7.62]
FRN %	33.8	41.7	[0,100]

Paper version

Table 3: Eligible Bonds Description			
	Mean	Sd	$[P_{10}, P_{90}]$
N Bonds	10.77	27.70	[1.00, 16.90]
Max maturity (years)	10.42	11.93	[2.00, 26.90]
Min maturity (years)	2.97	4.91	[0.00, 6.90]
Max coupon			
Min coupon			
FRN			

N = 186 auctions.

Replicated version

Figure 3: Table OS.1

9.3 In-Text Numbers

In-text numbers	Program	Line Number	Reproduced?
Section 2 Sample descriptives (N auctions, N bidders, NOI, etc.)	normalize_prices.m		No
Section 2 Price cap statistic	firststagebidding.m		Yes
Section 6.1 Status quo surplus bounds	calculate_surplusraw.m		No
Section 6.1 Truthful bidding surplus bounds	postestimation_clean.m		No
Section 6.2 Price bias (cents, %)	postestimation_clean.m		No
Section 6.3 Risk SD, hedging effectiveness	postestimation_clean.m		No
Section 7 CF surplus, price bias, hedging	postmain_cfs.m		No
Appendix C.2 Risk aversion comparison	riskaversion.m		No

In-text numbers	Program	Line Number	Reproduced?
Appendix C.5.1 Truthful reporting incentives	truthfulpimmcheck.m		No
Appendix C.5.1 Quote manipulation calibration	round1_quotescalibration.m		No
Appendix C.6 Customer orders robustness	robustnesschecks.m		No
Appendix C.7 Bid shading with positions	postestimation_clean.m		No
Appendix D CF robustness (volume caps, positions)	smc_cfs_yin.m		No

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☒ There are in-text numbers, but they are not identified in the code
- Identification in README is ambiguous.
- Section 2 in the case of Price cap statistic probably identifies to "Table 1 illustrates [...] there is an excess supply, of \$47.397M" on page 8 which is correctly stated from Table 1.
- On page 9 we have "[...] the IMM of 55.75 in the example" which matches with results in Table 2.
- The rest of in-text numbers have not been checked because the replication did not output the rest of tables and figures.

9.4 Other Issues

- Title of the Readme not formatted as in the guidelines.

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☒ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☒ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements